Tactile Device for Braille-Based Communication in Inclusive Education

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***Abstract*—This research aims to improve inclusive education for visually impaired people in Sinhala-speaking communities by introducing a prototype tactile Braille display. To enhance educational accessibility, the device converts digital text into Braille using an electromechanical process. Utilizing a Convolutional Neural Network (CNN) model, 97.6 % training accuracy, 82 % testing accuracy, and 0.88 precision were attained. A mobile app-based input technique showed better precision, 0.96, and accuracy (92.1 %). The device's dependability was demonstrated by the entire system performance, which displayed an accuracy of 91.8 ± 0.3 % and a precision of 0.93. By addressing educational disparities and fostering inclusivity in learning environments, this prototype offered visually impaired students access to education in their native Sinhala language using a customized learning tool.**

Keywords—braille, inclusive education, tactile communication, text-to-braille, image processing

I. Introduction

Students with visual impairments encounter several obstacles within the educational system. Conventional print-based materials and screen interfaces frequently hinder knowledge access and active participation in class activities. Braille display apparatus (BDAs) has become a valuable tool to help overcome these obstacles. By transforming written content into a tactile version, BDAs enable visually impaired students to engage with educational materials in a manner comparable to that of their sighted counterparts. With an emphasis on their effects on academic achievement, social inclusion in mainstream courses, and literacy development, this study investigated the potential of the tactile Braille device (TBD) to assist inclusive education.

Braille literacy is essential for visually impaired students since it facilitates academic success and information access [1]. By offering a dynamic platform for reading Braille, improving reading fluency, and encouraging a love of reading, TBDs play a vital part in this process. Using these devices, students can interact with digital content like their sighted peers by translating digital text into tactile Braille. TBDs guarantee visually impaired students fair access to educational materials in an increasingly digital world, encouraging digital literacy and closing the achievement gap [2]. TBDs are equipped with features that cater to a broad range of educational demands and facilitate Braille reading. Students can engage completely in class activities because they make completing tasks like taking notes, finishing projects, and accessing complex diagrams and multimedia content easier. TBDs can improve visually impaired students' academic achievement through equal access to resources and information. A more inclusive learning environment where all students, regardless of visual ability, have equal opportunities to achieve academically is fostered by incorporating BDAs into routine classroom instruction. Beyond raising academic performance, TBDs are also essential in promoting social participation in the classroom. TBDs eliminate the communication obstacles that might separate visually impaired pupils by giving them access to peers and teachers in real-time [3]. They encourage cooperation and teamwork, crucial contemporary social learning and education elements. These exchanges improve learning opportunities and foster the growth of social skills, confidence, and self-worth, which makes the classroom a more accepting and encouraging place.

This research aims to develop a TBD for inclusive education. We mainly discussed the TBD for the Sinhala Braille-based communications. Despite their apparent advantages, their effects in various educational settings have yet to be thoroughly studied. Instead of integrating BDAs into ordinary classrooms, existing studies frequently concentrate on their technical features or use in certain circumstances. By investigating the effects of BDAs on academic performance, social inclusion, and Braille literacy in conventional classroom settings, this study seeks to close this gap. To effectively educate visually impaired students, it is imperative to consider the complete scope of BDAs. Through this research, we can better understand the advantages and disadvantages of existing BDAs and design more affordable, effective, and user-friendly devices.

II. Current Trends and Developments

The progress made in assistive technologies for people with visual impairments indicates a deliberate attempt to improve communication and accessibility. Dela et al. [4] described an optical text-to-Braille converter intended to help visually impaired persons read printed items, a significant advancement in Braille and text conversion technologies. Their method incorporated a haptic device that used piezoelectric technology to translate text into Braille after digitizing it using an optical character recognition (OCR) engine. Building on this framework, Kumari et al. [5] presented an improved Braille system that used an image capture device, image processing methods, and OCR to transform text into Braille. The Raspberry Pi was in charge of

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this translation. Solenoids displayed the Braille code, enabling blind persons to view and interpret non-Braille text. Loconsole et al. [6] presented the BrailleCursor, a full-size, 40-cell, patent-pending refreshable Braille display to enhance these developments. This novel device used an electromagnetic actuator to rearrange a whole row of metal pins in Braille cells composed of passive pins, all with the push of a single actuated cursor. Park et al. [7] have further contributed to this field by addressing the difficulties associated with electronic document access for the visually impaired. Their suggested method of converting electronic documents to Braille was intended to improve translation precision and encourage the social engagement of people with visual impairments.

Aishwarya et al. [8] suggested a Braille pad to help blind people with communication gaps in interpersonal comprehension and communication. With the help of this technology, blind people interacted in Braille while sighted users conversed in English on computers and mobile devices. Similarly, Bono et al. [9] studied the negotiation of intersubjective understanding between Japanese finger Braille interpreters (FBIs) and a deafblind man (DBM). The DBM communicated using voicing and finger Braille, having co-invented the technology with his mother. The study focused on other-initiated repair (OIR) tactics to investigate and fill knowledge gaps.

Figiel [10], who interviewed experts in Poland, detailed the difficulties experienced by blind translators and interpreters. The research employed Pierre Bourdieu's theory of capitals to categorize the experiences of sighted and blind translators into three distinct periods: analog, transitional, and digital. This allowed the study to uncover growing differences in cultural capital. According to the research, these differences were made worse by inaccessible digital tools, which increased the likelihood that blind translators would be further excluded from the workforce.

Tab1. Summarize the current research technologies and their key features.

TABLE I. SUMMARY OF THE LITERATURE REVIEW

Ref	Technology Used	Key Features
[4]	Optical Text-to-Braille Converter	Uses piezoelectric haptic device; OCR engine digitizes text; translates to Braille.
[5]	Improved Braille System	Image capture device, image processing, OCR, and Raspberry Pi for text-to-Braille conversion.
[6]	BrailleCursor	40-cell refreshable Braille display using electromagnetic actuators.
[7]	Electronic Document to Braille Conversion	Method to enhance translation precision for electronic documents.
[8]	Braille Pad	Wireless technology, machine learning for Braille-to-text translation; enables communication with sighted users.
[9]	Japanese Finger Braille for Deafblind Individuals	Voicing and finger Braille communication; explored intersubjective understanding tactics.
[10]	Study on Blind Translators	Categorized experiences of blind translators across analog, transitional, and digital periods.

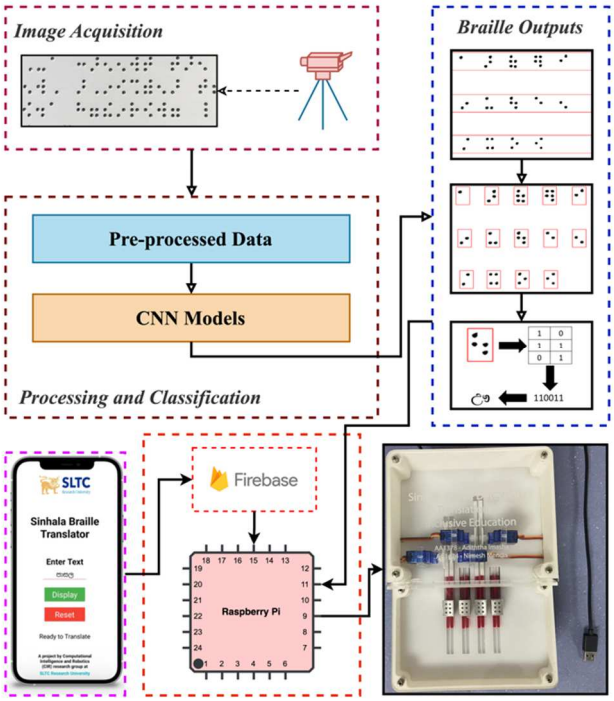


Fig. 1. The architecture of the proposed tactile Braille display system

Although there have been notable developments in assistive technologies for the blind and visually impaired, as shown in Tab. 1, such as the combination of tactile and machine learning (ML) for the detection of Braille text, no previous study has concentrated on creating a refreshable tactile display specially designed for Sinhala Braille. In inclusive education, where language-specific accessibility is crucial for fair learning opportunities, the lack of an integrated approach for Sinhala Braille devices reveals an essential gap.

This study intends to improve accessibility and inclusivity in education for visually impaired people by creating a refreshable Sinhala Braille tactile display, an essential tool for closing the communication and learning divide in Sinhala-speaking communities.

III. DEVELOPMENT OF BRAILLE INTERPRETER

This research aimed to develop a cost-effective Sinhala Braille display with essential functionality for visually impaired people. The proposed system's general block diagram, which shows the vital elements and workflow for this proposed strategy to assist visually impaired students, is shown in Fig. 1. The system's three main parts were a six-pin Braille cell, a slider mechanism, and a control system to operate the sliders.

As shown in Fig. 2, six individually operated pins were incorporated inside each Braille cell to offer tactile feedback for producing Braille characters. Rivet pins built to order were employed to satisfy the system's unique actuation requirements. The diameters of these rivet pegs vary slightly but not enough to compromise readability; they were made to resemble regular Braille dots. The semi-oval base of the rivets was designed to match the slider's curve, guaranteeing accurate and seamless movement. These personalized pins effectively replicate the raised dots needed for Braille reading.

The Braille cell dimensions were changed to improve the system's readability while staying near average letter size

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norms. This allowed a fingertip to cover the whole cell area. The dots in the suggested system were elevated and placed more prominently than in conventional Braille cells, and the cells were marginally more significant. These variations could be seen by comparing the dimensions: the system used a dot height of 0.8 mm and a dot diameter of 1.5 mm. With the system remaining as near to the standard Braille format as possible, some modifications were made to enhance tactile engagement. Aluminium was the material of choice for the Braille cells because of its strength and longevity. With this option, the cells could be used continuously without deteriorating.

Dual-diameter pinholes were used to construct the Braille cells to enable effective movement and control. The pinholes were 3 mm in diameter at the bottom to facilitate easy pin movement and 1.5 mm in diameter at the top to enable fine pin articulation. Every Braille cell had two sliders, each with three pins under its control. The system generated any possible pin combination for Braille representation with this configuration. The number of possible pin combinations, C, is given by the formula: $C=2^n$. Where n is the number of pins that each slider controls. The technique ensured the creation of Braille characters with $C=8$ unique pin combinations when $n=3$. The sliders were actuated using a rack and pinion mechanism. The linear displacement of the slider, d, was related to the rotation of the pinion gear by the following equation: $d=r_p \cdot \theta$. Where θ is the servo motor's rotation angle, and r_p is the pinion gear's pitch radius. The linear displacement was $\approx 15.7\ mm$ if the pitch radius was 10 mm and the rotation angle was $\theta=90^\circ$. In doing this, the sliders were positioned correctly to produce the appropriate pin combinations for Braille characters. Using the formula $D=N \cdot P$, where N is the number of teeth on the rack, and P is the gear teeth pitch, the rack length was optimized for adequate travel. To ensure sufficient movement for all eight pin combinations, the rack needed to move a minimum of 63 mm when using a 42-tooth pinion with a 1.5 mm tooth pitch.

The SG90 motors were used to rotate up to 180 degrees, and Pulse Width Modulation (PWM) was used to control the rack and pinion mechanism. The equation gives the relationship between the PWM signal and the servo motor's angle: $PWM = T_{min} + \theta \cdot ((T_{max}-T_{min})/\theta_{max})$. Where the maximum rotation angle, 180°, is denoted by θ_{max}, the maximum pulse width (T_{max}) is usually 2 ms, and the minimum pulse width, T_{min}, is usually 1 ms. The pulse width was 1.5 ms at a rotation angle of $\theta=90^\circ$. The servo motor was spined to the proper angle thanks to this exact control, which also moved the slider and modified the pins to produce Braille characters.

TABLE II. EXAMPLES OF SINHALA BRAILLE MAPPING FUNCTION

Sinhala Character	Braille Pattern	Code		PWM (ms)
		Binary	*Decimal*	
අ (a)	⠁	1	1	1.016
ආ (aa)	⠉	11	3	1.048
ඇ (ae)	⠋	101	9	1.143
ඈ (aae)	⠛	11001	25	1.397
ඉ (i)	⠑	100001	33	1.524
ඊ (ii)	⠓	111	7	1.111
උ (u)	⠥	101000	40	1.635
ක (ka)	⠇	1110	14	1.225

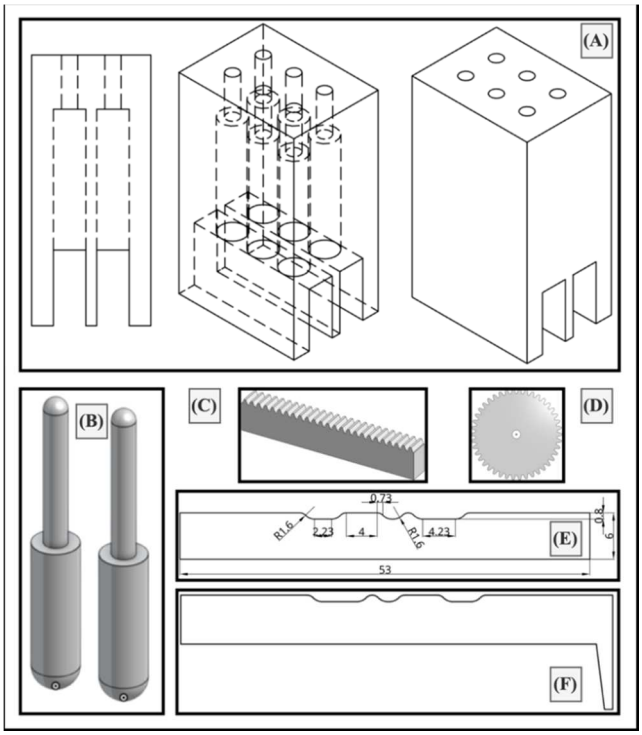


Fig. 2. Proposed Braille system mechanical structure, (A) Slider, (B) Rack of the slider, (C) Pinion of the slider, and (D) Braille cell

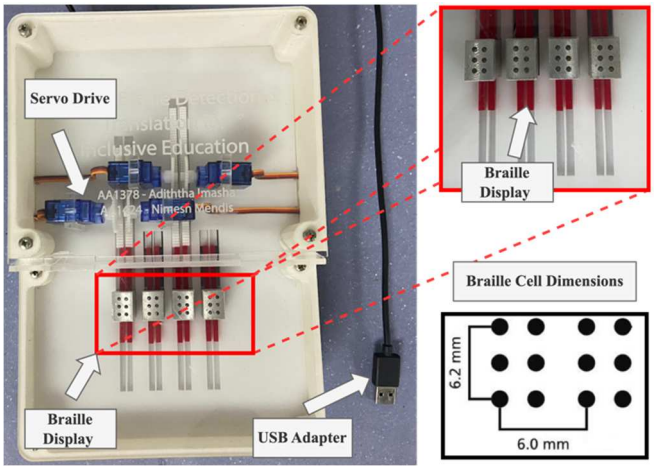


Fig. 3. Refreshable tactile Sinhala Braille display

The PCA9685 servo motor driver supported up to 16 servos concurrently and was responsible for the SG90 servo motors. Utilizing the I2C protocol, the PCA9685 driver lowered the number of GPIO pins the microcontroller needed. The servo motor angles were controlled by the Raspberry Pi microprocessor, which allowed the display to alternate between distinct Braille Sinhala characters. Six dots, each of which can be raised *(1)* or sunk *(0)*, make up a Braille cell. The universal equation for converting the dot pattern of a

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Braille cell to a decimal value can be described using the equation $B = \sum_{i=1}^{6} d_i \cdot 2^{i-1}$.

As shown in Tab. 2, the binary Braille patterns of the Sinhala characters were mapped to appropriate motor signal values to use PWM to operate the SG90 servo motors. Adjusting the pulse width changed the motor's angular position, directly regulating the Braille system's slider mechanism.

IV. Sinhala Text to Tactile Braille Display

Developing a Sinhala text-to-Braille conversion system is a significant step towards enabling visually impaired people in Sri Lanka, where Sinhala is the official language, to access written content. This research used a smartphone application as the user interface to process and translate Sinhala text into Braille code. The matching Braille characters were shown on a TBD (see Fig. 3) using affordable hardware parts like slider mechanisms and servo motors.

Upon entering the Sinhala text into the smartphone application, distinct Braille patterns were generated. A Braille cell's raised and sunken dots correspond to a binary code representing each Sinhala character. These binary patterns were necessary since they specify how the servo motors, which power the actual Braille display, should move. For example, the binary code 000001 represented the Sinhala letter "අ" (a) and corresponded to a specific arrangement of elevated pins in the Braille cell. The mobile app used a wireless connection to communicate these codes to the Braille display system to ensure a smooth and effective user-device interaction.

Six-pin Braille cells, with the pins organized in two columns of three, were the foundation of the Braille display system. Depending on the binary input, each pin was raised or sunk to represent the Braille characters visually. These pins were moved by a dual-slider mechanism, each of which can handle three pins. As a result, the device produced eight distinct pin combinations that corresponded to distinct Braille patterns for distinct Sinhala characters. The rack-and-pinion system, driven by SG90 servo motors, moved the sliders to obtain the appropriate pin configurations for every character.

The SG90 servo motors, which precisely controlled angular movements to control pin placements, were essential to the slider's operation. PWM signals were used in these motors to control the motor shaft's rotational angle. As seen in Fig. 4, the Braille cells were managed according to the PWM pulses produced. The angle θ was derived from the binary Braille code. For instance, for the letter "අ" (a), the binary pattern 000001 corresponded to a motor angle of $\theta=(1/63) \times 180=2.857^{\circ}$. The pulse width required to achieve this angle was: $PWM=1+ (2.857/180) \times(2-1)=1.0159$ *ms*. Similarly, for the letter "ආ" (aa), the binary code 000011 corresponded to a motor angle of $\theta=3/63\times180=8.571^{\circ}$.

A Raspberry Pi microprocessor controlled the system by processing data from the mobile app and producing the necessary PWM signals to operate the servo motors. The Raspberry Pi's interface with the PCA9685 driver ensured real-time performance and seamless communication through its GPIO ports. This configuration allowed the system to transition between several Braille characters smoothly, giving the user a responsive and effective interface. The ability to precisely manipulate pin designs and motor angles ensured that Braille translations of Sinhala text were correct, offering visually impaired people in Sri Lanka an accessible solution.

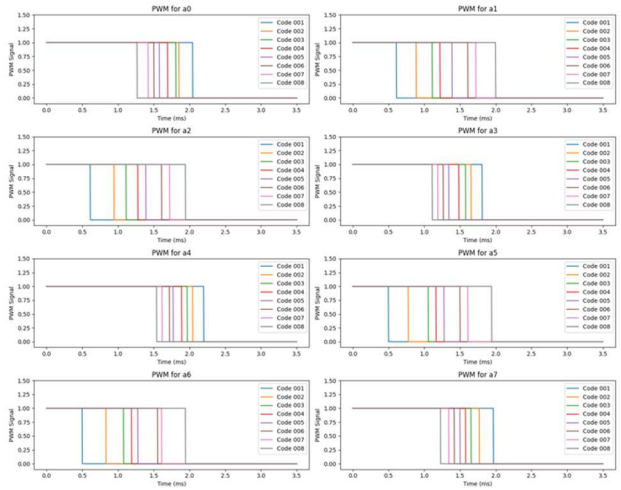


Fig. 4. PWM signals of the refreshable Braille servo mechanism

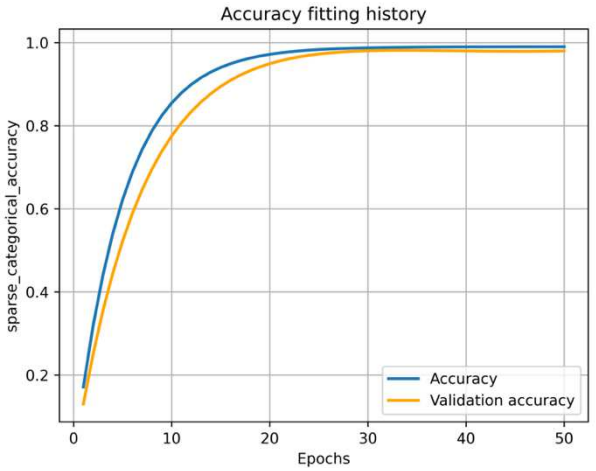


Fig. 5. CNN model's performance curve during the training process

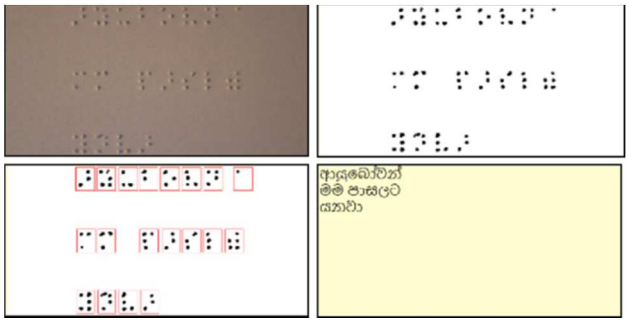


Fig. 6. Braille to Sinhala text conversion results of the ML model

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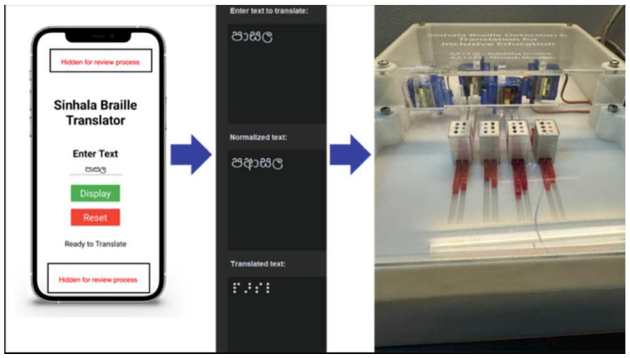


Fig. 7. Sinhala text to TBD conversion system

V. Braille Image to Tactile Braille Display

The capture of Braille documents was the initial step in the process of employing CNN to transform Braille images into a TBD. Braille slates with styluses, typewriters, or printers were used to make these documents. This research focused on the Braille slate and pen, which students in Sri Lanka commonly used to prepare documents. The Braille documents were embossed on regular white A4 paper using this technique. Although a flatbed scanner would have been the ideal tool, access to one was restricted, so instead, controlled lighting was used to take pictures with a smartphone camera.

Several preprocessing procedures were used after they were taken to prepare the image for recognition. To reduce the complexity of the image data, the photographs were first converted from RGB to greyscale. Subsequently, the Braille dots were preserved, but salt-and-pepper noise was eliminated using a median filter. Unlike other smoothing filters, the median filter preserved edges by substituting each pixel's median value for its neighboring pixels. To further minimize noise, a mean filter was applied after the median filter, yet this filter had the potential to blur the Braille characters. The mean filter replaced each pixel value with the average of its surrounding pixels.

Following thresholding, dilation was used to repair Braille dots that had partially worn away during the mean filtering procedure. Extending the Braille dots' bounds aided in regenerating their entire structure. Segmentation was breaking up the image into distinct Braille cells to be categorized. The image was divided into rows first, and then the Braille cells in each row were isolated. This process took two steps. Each cell was identified using the closely spaced Braille dots; the detected sections were then designated with rectangular boxes and stored for additional processing. Classifying the Braille cells came next after segmentation was finished. Six separate sections, or dots, made up a Braille character; the arrangement of the dots determined the character.

TABLE III. Binary Code and Column Code of the Word "School" in Sinhalse

Sinhala Character	Binary Code	Column Code			
		C1	*C2*	*C3*	*C4*
ප	111100	111	100	100	65
ළා	001110	001	110	10	100
ස	011100	011	100	105	60
ල	111000	111	000	70	125

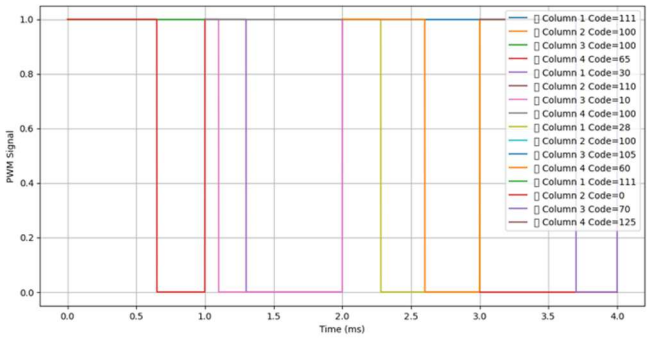


Fig. 8. PWM outputs of the refreshable servo mechanism for the Sinhala word "පාසල"

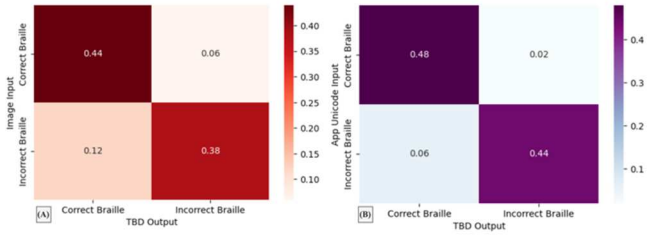


Fig. 9. Confusion matrix of the system (A) CNN model-based input, and (B) Sinhala Unicode-based input

TABLE IV. Accuracy of the Different Sinhalese Words Examples

Test #	Sinhala Word	Braille Cell Word	TBD Output	Accuracy	
				CNN Model	*Mobile App Model*
1	වචනය	වචන ය		81.3 %	89.4 %
2	ගිරය	ගඉර ය		80.7 %	88.9 %
3	ළමයා	ළමය ළා		80.9 %	90.4 %
4	පාසල	පළාස ල		81.5 %	91.2 %

Using the TensorFlow and Keras packages, a sequential model was constructed using the CNN method. The proposed CNN model started with a convolutional layer (32 filters) that extracted features before max-pooling was used to downsample the 28 × 28 input images. Feature extraction was improved with a second convolutional layer (64 filters), and max-pooling further reduced dimensions. Before the classification output layer, the data was compressed and processed through a dense layer (128 units). After 520 images from the combined dataset were enhanced using various manipulations like rotation, darkening, and shifting, 1,560 images were obtained. The dataset was split into 70 % for training, 20 % for validation, and 10 % for testing. The model's loss function was sparse categorical cross-entropy. To evaluate the correctness of the model, a wide range of Braille characters with different levels of complexity were used. The model's performance indicators were monitored during multiple training epochs, as seen in Fig. 5. High classification accuracy was shown by the primary metric, sparse categorical accuracy, which converged at 97.6 % after 40 epochs.

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VI. Results and Analysis

The findings and an analysis of the experimental protocol are discussed in this section. The system received inputs via an app-based interface or images. Therefore, the output findings were compared appropriately. Fifty test trials assessed the system's performance during the experimental procedure. The Sinhala Unicode input method and the Braille image acquisition were used in each trial.

For user convenience, the graphical user interface (GUI) showed the completed translation next to the identified Braille cells. This technique was demonstrated by translating a basic four-word Sinhala sentence, as seen in Fig. 6. The approach was more accurate and valuable for both instructors and Braille learners since it incorporates user input. The procedure for translating Sinhala Unicode text to Braille and transferring it to the TBD is shown in Fig. 7. The image shows how the text in Sinhala was converted to Braille and how each Braille cell's binary and column codes are produced. Tab. 3 provides a full breakdown of the binary codes for the Braille cells that correlate to the word "*School (පාසල)*." The motor control PWM signals based on the binary and column codes generated in the preceding step are shown in Fig. 8. According to the illustration, the motor controller received these PWM pulses, which corresponded to four column code sets, and used them to generate the Sinhala term "*School (පාසල)*."

The accuracy of Sinhala word identification in Braille using a CNN model and a mobile app model is contrasted in Tab. 4. In Test #1, the model accuracy achieved by the CNN was 81.3 %, but the mobile app model obtained 89.4 %. Similarly, Test #2 displayed an accuracy of 88.9 % using the app and 80.7 % using the CNN model. Test #3's accuracy with CNN was 80.9 %, whereas accuracy with the mobile app was 90.4 %. Test #4's accuracy with the CNN model was 81.5 %, while the mobile app was 91.2 %. Regarding Sinhala Braille word recognition, the mobile app model routinely beats the CNN model.

The system's confusion matrix is displayed in Fig. 9. The image-acquisition-based system had a precision of 0.88 and an accuracy of 82 %. On the other hand, the input testing of mobile apps using Sinhala Unicode yielded far higher results, with 92.1 % accuracy and 0.96 precision. These findings demonstrate the efficacy of the Sinhala Unicode-based mobile app for the particular application by showing that it outperforms the image acquisition-based system in terms of accuracy and precision. After 50 trials, the data were compared to determine the system's overall accuracy. The system attained a precision of 0.93 and 91.8 ± 0.3 % accuracy.

Twenty people with visual impairments were evaluated on the Sinhala words in Tab. 2. Of these, 16 people (80 %) with more Braille expertise correctly recognized the words. However, one participant (5 %) only managed to identify the word from Test #4, while three participants (15 %) had trouble recognizing words from Tests #2 and #3.

VII. Conclusion

For visually impaired students who speak Sinhala, a tactile Braille gadget represents a significant advancement in inclusive education. The prototype utilized a CNN to translate digital text into Braille electromechanically. Its results showed a precision of 0.88, testing accuracy of 82 %, and training accuracy of 97.6 %. Adding a mobile app-based input mechanism enhanced these measures to 92.1 % accuracy and 0.96 precision.

The device's accuracy and precision make educational resources more accessible to students who speak Sinhala and encourage diversity. On the other hand, drawbacks include device maintenance difficulties and problems with scaling to different languages. In inclusive education, the future of tactile Sinhala Braille displays depends on improving price, portability, and text-to-Braille conversion in real-time. By incorporating AI-powered translation tools, visually impaired students can benefit from increased accessibility and smooth cross-subject learning.

The development of proposed tactile devices brings up ethical and practical issues. Data privacy is crucial, particularly when devices have connectivity for accessing learning resources, necessitating strong security measures against unwanted access. Adoption is made more difficult by legal requirements like certifications and adherence to academic standards. Cooperative initiatives, financial aid, and expedited approval procedures are required to overcome these obstacles and allow for broad device availability.

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